Behavioural repertoires in JuPedSim

Model description of the rule-based entrance model, branch behavioural-repertoires (4c5b706d8)

30 October 2026

1 Scope

This note describes the model as implemented on the branch `behavioural-repertoires` at commit `4c5b706d8`. The library part lives in `jupedsim.repertoires`, and the entrance repertoires and rules in `jupedsim_examples.repertoires`. All numerical values are modelling assumptions; none is calibrated against data. The entry clearance of 0.6 m (Section 6.3) is an exploratory value.

The concept of behavioural repertoires follows Sieben and Postmes (2025). The rules, equations and values below are choices made for this model and are not taken from that paper.

2 Two levels of control

The operational model of JuPedSim (CFSM or CFSM V3) moves agents towards a target at a desired speed while avoiding collisions. A *behavioural layer* above it decides which behaviour each agent engages in and turns that behaviour into the operational model's two inputs: the target $\mathbf{x}_i^*$ and the desired speed v_i^0.

Two quantities describe an agent's behaviour:

- the **repertoire** (queueing or overtaking), chosen by a *selector*; it fixes the social anchor and the permitted manoeuvre;
- the **execution mode** (standing, advancing, passing), set by the repertoire at every control step without a new selection.

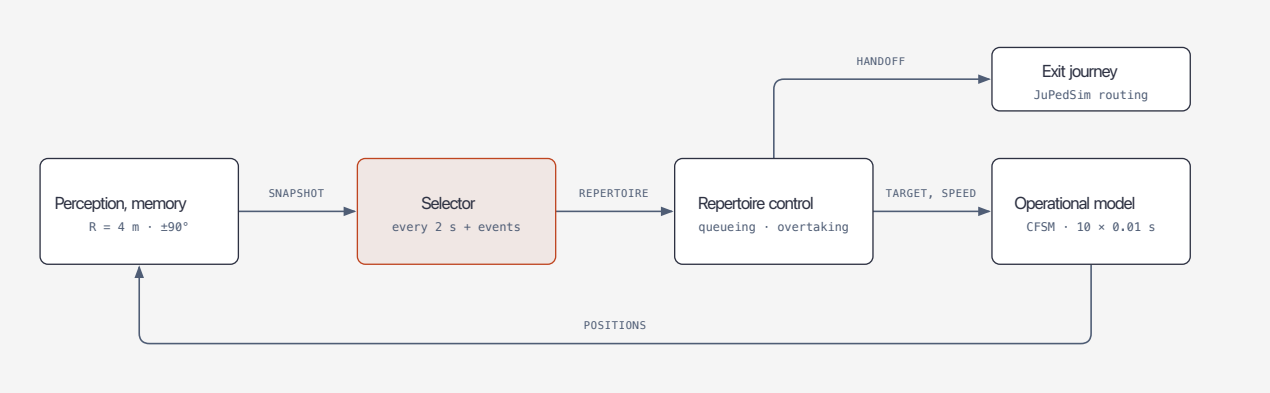


Figure 1: One control interval. The selector (accent) is the only component that can be replaced, e.g. by a learned or external decision model.

The layer acts every $n_c = 10$ iterations of the operational model, i.e. every $\Delta t_c = n_c\,\Delta t = 0.1$ s with $\Delta t = 0.01$

s. Agents near the entrance are handed off to an ordinary JuPedSim exit journey and leave the layer (Section 7).

3 Progress and perception

Let E be the entrance and $\mathbf{x}_i(t)$ the position of agent i. Progress and the unit direction towards E are

$$p_i(t) = -\|\mathbf{x}_i(t) - E\|, \qquad \mathbf{u}(\mathbf{x}) = \frac{E - \mathbf{x}}{\|E - \mathbf{x}\|}.$$

Progress increases as the agent approaches E.

Agent i perceives agent $j \neq i$ if j is within the perception radius R and inside the view cone of half-angle φ around the heading $\mathbf{h}_i = \mathbf{u}(\mathbf{x}_i)$:

$$j \in \mathcal{P}_i \iff \|\mathbf{r}_{ij}\| \le R \ \wedge \ \big(\mathbf{r}_{ij} \cdot \mathbf{h}_i \ge \cos\varphi \, \|\mathbf{r}_{ij}\| \ \vee \ j = \tau_i\big), \qquad \mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i,$$

with $R = 4$ m and $\varphi = 90°$, so the cone is the half-plane in front of the agent. The term $j = \tau_i$ applies only while i is overtaking: τ_i is the agent being passed. It stays perceived within R once it falls behind, so the overtaker can tell when the pass is complete. Without this exception the target leaves the half-plane before the completion margin of Section 6.4 is reached. Queueing agents have no tracked agent.

4 Memory

Each agent keeps, for a window of $W = 8$ s, its progress history and the passes it has observed.

Progress gained over a span T is

$$\Delta p_i(t; T) = p_i(t) - p_i(t_0), \qquad t_0 = \min\{t_k \ge t - T\},$$

over the recorded control times t_k. If the history does not reach back to $t - T$, $\Delta p_i = +\infty$, so a newly controlled agent is never counted as stalled.

Passes. Agent j passes i at time t if j was perceived behind i, $p_j - p_i < -\delta$, at some time within the window, has been perceived at every control step since, and now satisfies $p_j - p_i > \delta$, with $\delta = 0.2$ m. The number of passes since $t - T$ is $N_i(t; T)$.

Anchor lost is a flag set when the agent's anchor has left the simulation.

5 Selection

5.1 When decisions are taken

An agent requests a decision when one is periodically due or when an event has occurred:

- *periodic*: every $T_d = 2$ s; agent k, numbered in order of entry into the layer, is due at $t = \frac{k \bmod 10}{10} T_d + m\,T_d$ for $m = 0, 1, \ldots$, which spreads decisions over the period;
- *events*: the event emitted by the active repertoire at the previous step (completion or abandonment), a pass observed at this step, or the anchor-lost flag.

All requests of one control step are built from the same snapshot of perception and memory and are sent to the selector in one batch. The selector's answers are applied together.

5.2 Rule-based selector

Incentive to overtake is

$$I_i = \left[\Delta p_i(t; T_s) < \Delta p_s\right] \ \vee \ \left[N_i(t; T_n) \geq n_p\right],$$

with $T_s = 4$ s, $\Delta p_s = 0.3$ m, $T_n = 6$ s and $n_p = 2$: the agent is stalled, or it has been passed repeatedly.

Feasibility is a passing point beside the anchor that satisfies the entry clearance (Section 6.3):

$$G_i = \left[\pi_i \in \mathcal{P}_i\right] \ \wedge \ \mathcal{C}_i(r_{\text{in}}) \neq \varnothing.$$

The transitions are:

Current repertoire	Condition	New repertoire	Logged reason
queueing	$I_i \wedge G_i$	overtaking	
queueing	$I_i \wedge \neg G_i$	queueing	`overtake_withheld_no_gap`
queueing	$\neg I_i$	queueing	
overtaking	event *completed* or *abandoned*	queueing	
overtaking	otherwise	overtaking	

The incentive is not stored. It holds as long as the memory supports it, so an agent with an incentive enters overtaking as soon as a gap opens within that time.

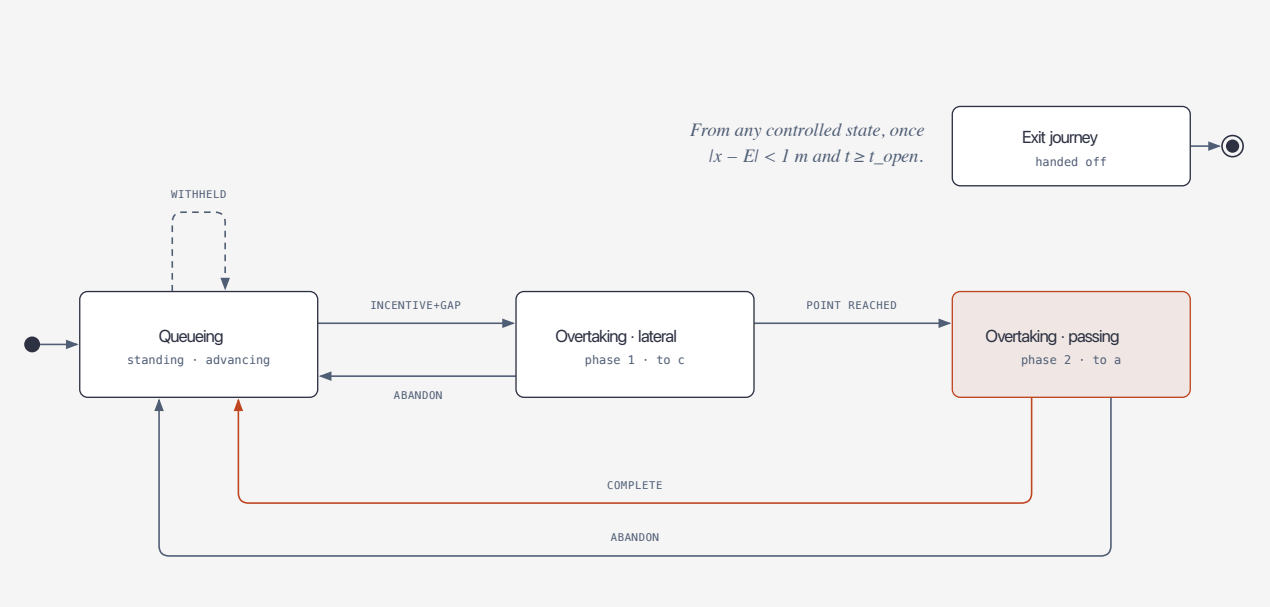


Figure 2: Repertoire and phase transitions of one agent. Abandonment from the lateral phase has three causes (no gap, target lost, stalled); from the passing phase, two (target lost, stalled).

6 Repertoires

6.1 Queueing

The **predecessor** is the nearest perceived agent ahead by more than the margin $\delta_a = 0.2$ m:

$$\pi_i = \underset{j \in \mathcal{P}_i,\, p_j > p_i + \delta_a}{\arg\min} \|\mathbf{x}_j - \mathbf{x}_i\|,$$

with ties broken by the smaller id. The agent keeps its predecessor until the predecessor is lost, no longer perceived or no longer ahead by δ_a, and then acquires a new one. The predecessor is the agent's own belief; no global queue order exists.

The **queue target** lies at spacing s behind the predecessor, along the predecessor's direction to the entrance:

$$\mathbf{q}_i = \begin{cases} \mathbf{x}_{\pi_i} - s\,\mathbf{u}(\mathbf{x}_{\pi_i}) & \text{if } \pi_i \text{ exists,} \\ E & \text{otherwise,} \end{cases} \qquad s = 0.7 \text{ m}.$$

The desired speed is $v_q = 1.0$ m/s. The execution mode is *standing* if $\|\mathbf{x}_i - \mathbf{q}_i\| < 0.15$ m and *advancing* otherwise.

6.2 Overtaking

On entry the anchor becomes the passing target τ_i, and the manoeuvre starts in phase 1. The desired speed is $v_o = 1.4$ m/s.

6.3 Passing points

With the normal $\mathbf{n} = (-u_y, u_x)$ of $\mathbf{u} = \mathbf{u}(\mathbf{x}_\tau)$, the two lateral candidates beside the target are

$$\mathbf{c}^{\pm} = \mathbf{x}_\tau \pm d\,\mathbf{n}, \qquad d = 0.8 \text{ m}.$$

A candidate is feasible with clearance r if it lies in the walkable area $\mathcal{W}$ shrunk by the wall clearance w and no perceived agent other than the target is closer than r:

$$\mathcal{C}_i(r) = \left\{\mathbf{c} \in \{\mathbf{c}^+, \mathbf{c}^-\} : \mathbf{c} \in \mathcal{W} \ominus B_w \ \wedge\ \|\mathbf{c} - \mathbf{x}_j\| \geq r \ \ \forall j \in \mathcal{P}_i \setminus \{\tau_i\}\right\}, \qquad w = 0.25 \text{ m}.$$

The chosen point is the feasible candidate closest to the agent, ties going to $\mathbf{c}^+$: $\mathbf{c}_i^* = \arg\min_{\mathbf{c} \in \mathcal{C}_i(r)} \|\mathbf{x}_i - \mathbf{c}\|$.

Starting needs more room than continuing. The clearance is

$$r = \begin{cases} r_{\text{in}} = 0.6 \text{ m} & \text{at entry and on the first control step,} \\ r_{\text{cont}} = 0.4 \text{ m} & \text{afterwards.} \end{cases}$$

The first control step shares the snapshot of the entry decision, so the agent steers to the point that justified the entry and not to a nearer point that only meets r_{cont}. Overtakes started with a clearance just above a single threshold of 0.4 m were otherwise ended within a few control steps by small sideways movements of neighbours (Section 10).

6.4 Execution and termination of an overtake

At every control step the overtaking repertoire checks, in this order:

1. **Handoff** (Section 7).
2. **Target lost**: $\tau_i \notin \mathcal{P}_i \Rightarrow$ abandon (`overtake_abandoned_target_lost`).
3. **Completion**: $p_i > p_\tau + m_c$ with $m_c = 0.3$ m $\Rightarrow$ event `overtake_completed`; target E, mode *advancing*.
4. **Stall**: $t - t_{\text{commit}} \geq T_a$ and $\Delta p_i(t; T_a) < \Delta p_a$, with $T_a = 2$ s and $\Delta p_a = 0.1$ m $\Rightarrow$ abandon (`overtake_abandoned_stalled`).
5. **Phase 1** (lateral): if $\mathcal{C}_i(r) = \varnothing$, abandon (`overtake_abandoned_no_gap`). If $\|\mathbf{x}_i - \mathbf{c}_i^*\| < 0.3$ m, switch to phase 2. Otherwise steer to $\mathbf{c}_i^*$.
6. **Phase 2** (passing): steer to the ahead point $\mathbf{a}_i = \mathbf{x}_\tau + \ell\,\mathbf{u}(\mathbf{x}_\tau)$, $\ell = 0.8$ m.

Abandoning sets the target to the agent's own position, the desired speed to v_q and the mode to *standing*. Losing the target is never counted as completion. A completion or abandonment event triggers a decision at the next control step, where the selector returns the agent to queueing.

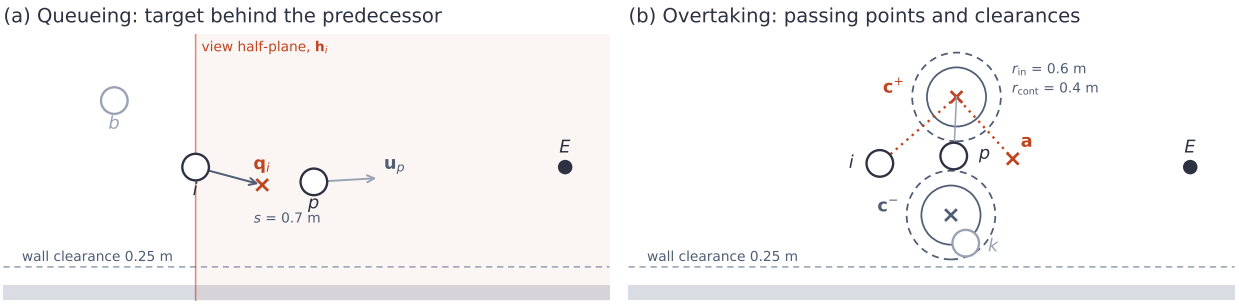


Figure 3: Geometry. (a) Queue target $\mathbf{q}_i$ at $s = 0.7$ m behind predecessor p; agent b is behind i's view half-plane and is not perceived. (b) Lateral candidates $\mathbf{c}^{\pm}$ at $d = 0.8$ m beside target p, with entry (dashed) and continuation (solid) clearance; k blocks $\mathbf{c}^-$, so i moves to $\mathbf{c}^+$ and then to the ahead point $\mathbf{a}$.

7 Handoff

An agent leaves the layer when

$$\|\mathbf{x}_i - E\| < r_h \quad \text{and} \quad t \geq t_{\text{open}}, \qquad r_h = 1 \text{ m},$$

from either repertoire and in any phase. It is switched to the exit journey and is then routed by JuPedSim alone. Handoff is not a completed overtake. The door opening time t_{open} is 0 in the entrance and demo scenarios and 90.5 s in the 2C120 scenario, where the door is closed until then.

8 One control step

1. Release agents that have left the simulation; set the anchor-lost flag of agents whose anchor has left.
2. Observe every controlled agent (perception and memory) at the same time t.
3. Enter the initial repertoire for agents added since the last step.
4. Collect decision requests (periodic or event-triggered) and pass them to the selector in one batch.
5. Apply all decisions; an agent whose repertoire changes enters the new one with $t_{\text{commit}} = t$.
6. Run the active repertoire of every agent: set its target and desired speed, or hand it off.
7. Log diagnostics and advance the operational model by n_c iterations.

A target outside the walkable area is not applied: the agent holds its position and its mode becomes *standing*.

9 Parameters

Symbol	Meaning	Value	Code name
Δt, n_c	time step; iterations per control step	0.01 s; 10	`dt`, `control_interval`
T_d	decision period	2 s	`decision_period`
R, φ	perception radius; half-angle	4 m; 90°	`radius`, `half_angle_deg`
W, δ	memory window; pass tolerance	8 s; 0.2 m	`window`, `pass_tolerance`

Symbol	Meaning	Value	Code name
T_s, Δp_s	stall span; minimum progress	4 s; 0.3 m	`t_stall`, `stall_progress`
T_n, n_p	pass span; passes needed	6 s; 2	`t_passes`, `n_passes`
δ_a	"ahead" margin	0.2 m	`ahead_margin`
s	queue spacing	0.7 m	`spacing`
v_q, v_o	desired speed queueing; overtaking	1.0; 1.4 m/s	`queue_speed`, `overtake_speed`
d	lateral offset	0.8 m	`lateral_offset`
w	wall clearance	0.25 m	`wall_clearance`
r_{in}, r_{cont}	entry; continuation clearance	0.6; 0.4 m	`entry_free_radius`, `continuation_free_radius`
—	lateral point reached	0.3 m	`lateral_reach`
ℓ	ahead offset	0.8 m	`ahead_offset`
m_c	completion margin	0.3 m	`complete_margin`
T_a, Δp_a	overtake stall span; minimum progress	2 s; 0.1 m	`abandon_span`, `abandon_progress`
—	standing tolerance	0.15 m	`standing_tol`
r_h	handoff radius	1 m	`handoff_radius`
t_{open}	door opening time	0 s	`door_open_time`

The 2C120 scenario uses $v_q = 2.5$ m/s and $v_o = 3.5$ m/s: the median of the participants' 90th-percentile speeds over the first 20 s is 2.5 m/s.

10 What the simulations show so far

Entrance scenario (20 m × 10 m approach area, 1 m wide entrance), CFSM, 90 agents, seeds 1–3, continuation clearance 0.4 m:

Entry clearance	0.4 m	0.5 m	0.6 m
Overtakes started	639	299	215
Completed	209	149	134
Abandoned: no gap / stalled / target lost	368 / 45 / 3	100 / 39 / 2	57 / 18 / 1
Abandoned within 0.5 s	198	25	7
Completed per run	55, 66, 88	54, 40, 55	48, 42, 44
Final exit time per run (s)	102.3, 109.4, 106.0	103.5, 107.2, 109.0	102.7, 108.4, 103.8

Before the target was tracked outside the view cone, no overtake completed (0 of 627 in the same runs); successful passes were logged as target loss.

Open points:

- In the 2C120 scenario with CFSM V3 and the rule-based selector, agents jam in front of the door after it opens (48 of 87 remain at 200 s, measured before the entry clearance was raised to 0.6 m). With CFSM, or with queueing only, the corridor empties. The cause is not yet identified.

- The labels *completed* and *abandoned* describe the controller, not observed behaviour. Whether the trajectories show queueing and overtaking has to be established by coding them as the experiments were coded.
- The rule-based selector can be replaced by another selector (for example one estimating a Jev-like value) without changing the repertoires.

11 Reference

Sieben, A. and Postmes, T. (2025). Behavioural repertoires in moving crowds: an observational approach. *Royal Society Open Science* 12, 241561. doi:10.1098/rsos.241561.